Lesson 12: Hearing It Happen

GameMaker

VIDEO GAME CREATION

This week: your game finally sounds like a game. Four moments - firing, hitting, clearing a wave, and losing - each get their own distinct cue.

Open your own [YourName]Game project from Lesson 11 - don't start a new one.

If you already built an optional sound idea in an earlier lesson (Lessons 2, 3, 4, or 6), keep it - today reuses it directly instead of replacing it. If you didn't, today's steps are written as if starting fresh.

Setup

1. Open your [YourName]Game project from Lesson 11.
2. Create three new Sound resources: sfx_wave and sfx_lose (and sfx_win, if you don't already have it from Lesson 6).

Level 1 - Minimal (MVP)

Four sounds fire at the correct moment, at least once each: a shot, a hit/explosion, a wave-clear (on any non-final wave), and a game over. No silence, no obviously wrong triggers.

3. Confirm sfx_shoot already plays via audio_play_sound() in your shoot input code (Lesson 3) - add it now if it's missing.
4. Confirm sfx_hit already plays via audio_play_sound() in obj_bullet's Collision event (Lesson 4/5) - add it now if it's missing, right after the score/enemies_remaining updates.
5. In obj_bullet's Collision event's nested if (from Lesson 11), add audio_play_sound(sfx_win, 1, false) in the true-win branch and audio_play_sound(sfx_wave, 1, false) in the else branch.
6. In obj_enemy's Step event's escape branch (where lives decrements), add audio_play_sound(sfx_lose, 1, false) inside the lives <= 0 check.
7. Press F5 (Run) and play through a full round - shoot, hit, an early wave clear, and a loss should each sound distinct.

```
// obj_bullet Collision event (revised):
obj_spawner.enemies_remaining -= 1;
obj_spawner.score += 10;
audio_play_sound(sfx_hit, 1, false);
if (obj_spawner.enemies_remaining <= 0) {
    if (obj_spawner.wave >= obj_spawner.total_waves) {
        obj_spawner.game_active = false;
```

```
        obj_spawner.has_won = true;
        audio_play_sound(sfx_win, 1, false);
    } else {
        obj_spawner.wave += 1;
        obj_spawner.enemies_remaining = 5;
        audio_play_sound(sfx_wave, 1, false);
        obj_spawner.spawn_wave();
    }
}
instance_destroy();

// obj_enemy Step event, escape branch (revised):
obj_spawner.lives -= 1;
if (obj_spawner.lives <= 0) {
    obj_spawner.game_active = false;
    obj_spawner.has_lost = true;
    audio_play_sound(sfx_lose, 1, false);
}
```

Alternate Path: GML Visual (no typing)

The same sound calls work as drag-and-drop blocks instead of typed code.

8. On obj_bullet's Collision event (GML Visual), add an "Audio Play Sound" block (sfx_hit) right after your existing score/enemies_remaining Variable blocks.
9. Add two more "Audio Play Sound" blocks to the win-check's If Variable block from Lesson 11 - sfx_win on the True output, sfx_wave on the False output.
10. On obj_enemy's escape branch, add an "Audio Play Sound" block (sfx_lose) inside the lives-reaches-zero If Variable block.

Level 2 - Complete (Alpha/Beta)

All four sounds fire reliably every time, with no missing or duplicate playback, and each is clearly distinguishable from the others by ear.

- Play a full round: shoot, hit, an early wave clear, and a loss each produce a distinct sound.
- If you built the optional win sound, confirm it still plays on the true win, and sfx_wave doesn't also fire on that same frame.
- audio_play_sound() calls layer naturally - confirm rapid shooting and a hit landing together both play correctly, with no sound cutting another off.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Audio: if you haven't already, add the optional win sound now - it slots into the exact same audio_play_sound() pattern as today's wave/lose sounds, just a third resource.
- Audio: add slight pitch variation to hits with audio_play_sound(sfx_hit, 1, false) followed by audio_sound_pitch(inst, random_range(0.9, 1.1)), where inst is the sound's returned instance handle.
- Structure: group all your Sound resources under one folder in the Asset Browser for organization, now that there are five of them.

If Something's Wrong

- No sound on wave-clear: confirm the sfx_wave call is actually inside the else branch, not after the if/else.
- Lose sound plays repeatedly: confirm the lose check is still guarded by "if (obj_spawner.lives <= 0)" as a one-time check (from Lesson 8), not re-triggering every Step.
- Wrong instance's variables referenced: confirm the obj_spawner. prefix is present throughout the revised blocks, since this code still runs from obj_bullet and obj_enemy.

Want More?

Student Resource List > GameMaker section, or manual.gamemaker.io